MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T24E1
METAL PARTS

1. SCOPE

1.1 This specification covers the metal parts for one type of detonator designated as Detonator, Electric, T24E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111 - Tape, Paper, Gummed (Sealing and Securing).
PPP-B-636 - Boxes, Fiber.

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipments and Storage.

DRAWINGS

ORDNANCE CORPS

P-86242 - Detonator, Electric, T24E1. Assembly and Details.

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(see 6.1) - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) P-86242, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be free of chips, dirt, rust, grease, solder splash, flux, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

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4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2.2 and 4.3, and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analyses on all material procured directly by the contractor.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and

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other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance. -Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection. -Physical inspection by the Government of contractor-accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies. -Deviations from the prescribed or agreed-upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions. -

4.2.1 Lot formation. -A lot shall consist of parts produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number.

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4.2.2 Examination. -Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.3.

4.2.2.1 Cup (see dwg. P-86242). -

Categories	Defects	Method of inspection
Critical:		
	1. Split, cut or crack through cup	Visual
Major:	AQL 0.65 percent	
	101. Thickness through bottom, maximum (max.)	Gage
	102. Inside diameter, minimum (min.)	Gage
	103. Sidewall thickness, min.	Gage
	104. Cup dented or distorted	Visual
	105. Any scratch or groove	Visual
Minor:	AQL 1.50 percent	
	201. Profile of lip	Visual
	202. Radius missing	Visual
	203. Finish improper	Visual
	204. Evidence of poor workmanship (see 3.3)	Visual

4.2.2.2 Plug and wire assembly (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
	101. Pull on lead wires	Test (see 4.3.1)
	102. Resistance between lead wires, min.	Test (see 4.3.2)
	103. Distance between shorter wires at bridge end of assembly, max.	Gage
	104. Large outside diameter of plug, min.	Gage
	105. Distance between ferrule side of plug and cup crimp surface	Gage
	106. Height of small diameter above cup crimp surface	Gage
	107. Plug cracked or chipped	Visual

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Categories	Defects	Method of inspection
Minor:	AQL 1.50 percent	
201.	Diameter of plug at end of taper, ...	Gage
202.	Length of lead wire at bridge end, min.	Gage
203.	Evidence of poor workmanship (see 3.3)	Visual
204.	Finish improper	Visual

4.2.2.3 Ferrule (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Outside diameter, max.	Gage
Minor:	AQL 0.40 percent	
201.	Length, min.	Gage

4.2.2.4 Carton, prior to sealing (see para's 5.1.1.1 and 5.1.1.2).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Incorrect number of rosettes	Visual
102.	Incorrect number of cups	Visual
103.	Incorrect number of assemblies in rosettes	Visual
Minor:	AQL 0.40 percent	
201.	Wire insecure	Visual

4.2.2.5 Carton, sealed (see 5.3).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least 1/2 inch around entire carton.	Visual
Minor:	AQL 1.00 percent	
201.	Marking misleading or unidentifiable	Visual
202.	Contents loose when carton is shaken	Manual

4.2.3 Inspection equipment. -For the performance of the examinations and tests specified in 4.2.2 and 4.3, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment identified by Index of Inspection Equipment No. (see 6.1), or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring, and testing equipment of the required accuracy and precision and that the instruments are of the proper type and

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range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures. -

4.3.1 Lead wire pull. -The lead wires of the plug assembly shall be subjected to the minimum axial pull specified on the applicable drawing. Observation shall be made for any indication of slippage of the lead wires within the plug.

4.3.2 Resistance between the lead wires of the plug. -A minimum voltage of 325 volts direct current shall be applied to the lead wires using a specified test circuit. The electrical resistance shall be measured to determine compliance with the requirements specified on the drawing.

5. PREPARATION FOR DELIVERY

5.1 Packing. -

5.1.1 Level C. -Detonator metal parts shall be packed for shipment as described below. The construction of the cartons and the material requirements of the carton shall comply with Federal Specification PPP-B-636. When assembled, the corrugations of the cartons and partitions shall parallel the carton depth. The manufacturer's joint on the carton shall be metal stapled or stitiched. Spacing of staples or stitches shall not average over 1 1/2 inches apart, center to center. A manufacturing tolerance of plus or minus 1/16 inch is permissible. The parts manufacturer shall be furnished preproduction samples of all packing materials so that proper fit of packing materials and metal parts can be determined. Minor modifications of packing material dimensions to provide a proper fit of all parts, as determined to be necessary by examination of sample packs, are permissible. All container voids shall be filled with dunnage to insure tight packing.

5.1.1.1 Plug and wire assembly (P-86242G). -Ten thousand plug and wire assemblies shall be packed in 40 rosettes of 250 each and secured by wrapping one wire around the extended sheaf of 249 wires. The 40 rosettes shall be packed in bulk in a 275 pound (lb.) test, RSC Style, "B" flute, corrugated fiberboard carton. The inside dimensions of the carton shall be 20 by 18 by 10 inches.

5.1.1.2 Cup (P-86242L). -Ten thousand cups shall be packed in bulk in a 275 lb. test, RSC Style, "B" flute, corrugated fiberboard carton. The inside dimensions of the carton shall be 10 by 10 by 5 inches.

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5.1.1.3 Ferrule (P-86242M). -Ten thousand ferrules shall be packed in bulk in a RSC Style 275 lb. test, double-faced corrugated fiberboard carton. The inside dimensions of the carton shall be: 4 by 4 by 2 1/4 inches.

5.2 Percentage of extra parts packaging. -

5.2.1 When feasible, for small and light weight parts, the entire packing lot quantity, including the percentage of extra parts, shall be packed in the prime container.

5.3 Sealing. -Carton flaps shall be secured with gummed Kraft paper not less than three inches wide, complying with Federal Specification UU-T-111 for Class 2 and Class 3 tape. The tape shall be centered the full length of the outer flap seams, and extend over each end of the carton at least three inches. The flap edge seams shall be similarly sealed with the tape extending at least three inches along the corners of the box. Tape shall be properly moistened and sufficiently compressed to insure tight adhesion to the carton. Cartons weighing up to 50 lbs. shall be sealed with Class 2 tape. Cartons weighing 50 lbs. or more shall be sealed with Class 3 tape. Alternatively, when acceptable to the installation using the metal parts, flaps may be glued using a suitable adhesive applied over at least 75 percent of the contact area between inner and outer flaps. When flaps are pulled apart after gluing, they shall show complete fiber failure in the glued area.

5.4 Marking. -Marking of the packing shall comply with Specification MIL-STD-129.

5.5 Shipping. -When component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

5.6 Data cards. -Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Inspection equipment. -Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.

6.2 Ordering data. -Procurement documents should specify the title, number, and date of this specification.

Notice. -When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, speci-

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